Experiment No.: 3 Name: Rohan Laharwani PRN: 24070521055

1. Experiment Title:

Implement control structures and form validation; create a grading system based on user-entered marks.

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, Javascript

3. Experiment Program Code:

Experiment 3.html

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Grading System</title>
    <link rel="stylesheet" href="Experiment 3.css">
</head>

<body>
    <div class="container">
        <div class="card">
            <h2>Student Grading System</h2>
            <form id="gradingForm">
                <div class="form-group">
                    <label for="name">Student Name</label>
                    <input type="text" id="name" placeholder="Enter student name">
                </div>

                <div class="form-group">
                    <label for="marks">Marks</label>
                    <input type="number" id="marks" placeholder="Enter marks (0-100)" min="0" max="100">
                </div>

                <input type="button" class="btn" value="Calculate Grade" onclick="gradeSystem()">
            </form>

            <div id="report"></div>
        </div>

        <footer style="text-align: center; margin-top: 30px; font-size: 14px; color: gray; border-top: 1px
solid lightgray; padding-top: 15px;">
            Made by Rohan Laharwani PRN - 24070521055
        </footer>
    </div>

    <script src="Experiment 3.js"></script>
</body>

</html>
```

Experiment 3.css

```
body {
    background-color: whitesmoke;
    color: black;
    font-family: sans-serif;
    margin: 0;
    padding: 0;
    display: flex;
    justify-content: center;
    align-items: center;
    min-height: 100vh;
}

.container {
    width: 100%;
    max-width: 400px;
    padding: 20px;
}

.card {
    background-color: white;
    padding: 30px;
    border: 1px solid lightgray;
    border-radius: 12px;
    box-shadow: 0 4px 12px lightgray;
}

h2 {
    color: darkblue;
    text-align: center;
    margin-top: 0;
    margin-bottom: 20px;
}

hr {
    border: 0;
    height: 1px;
    background-color: lightgray;
    margin: 20px 0;
}

form {
    display: flex;
    flex-direction: column;
    gap: 15px;
}

.form-group {
    display: flex;
    flex-direction: column;
    gap: 5px;
}

label {
    font-weight: bold;
    font-size: 14px;
    color: dimgray;
}

input[type="text"],
input[type="number"] {
    padding: 10px;
    border: 1px solid lightgray;
    border-radius: 6px;
    font-size: 16px;
}

input[type="text"]:focus,
input[type="number"]:focus {
```

```
    border-color: blue;
    outline: none;
}

.btn {
    padding: 12px;
    background-color: royalblue;
    color: white;
    border: none;
    border-radius: 6px;
    font-size: 16px;
    font-weight: bold;
    cursor: pointer;
    transition: background-color 0.2s;
}

.btn:hover {
    background-color: blue;
}

.btn-secondary {
    padding: 12px;
    background-color: darkgray;
    color: white;
    border: none;
    border-radius: 6px;
    font-size: 16px;
    font-weight: bold;
    cursor: pointer;
    width: 100%;
    margin-top: 20px;
}

.btn-secondary:hover {
    background-color: gray;
}

#report {
    display: none;
    margin-top: 20px;
}

#report.visible {
    display: block;
}

.report-details {
    display: flex;
    flex-direction: column;
    gap: 10px;
}

.report-details p {
    display: flex;
    justify-content: space-between;
    margin: 0;
    font-size: 16px;
}

.badge {
    font-weight: bold;
}

.status-pass {
    color: green;
}

.status-fail {
    color: red;
```

```
}

.grade-aplus, .grade-a, .grade-b, .grade-c, .grade-d, .grade-e {
    color: green;
}

.grade-f {
    color: red;
}

.report-view {
    background-color: whitesmoke;
    color: black;
    font-family: sans-serif;
    margin: 0;
    padding: 0;
    display: flex;
    justify-content: center;
    align-items: center;
    min-height: 100vh;
}
```

Experiment 3.js

```
function gradeSystem() {
    var name = document.getElementById("name").value.trim();
    var marks = document.getElementById("marks").value;

    if (name == "") {
        alert("Please enter student name: ");
        return;
    }

    if (marks == "") {
        alert("Please enter marks: ");
        return;
    }
    marks = Number(marks);

    if (marks < 0 || marks > 100) {
        alert("Marks should be between 0 and 100.");
        return;
    }

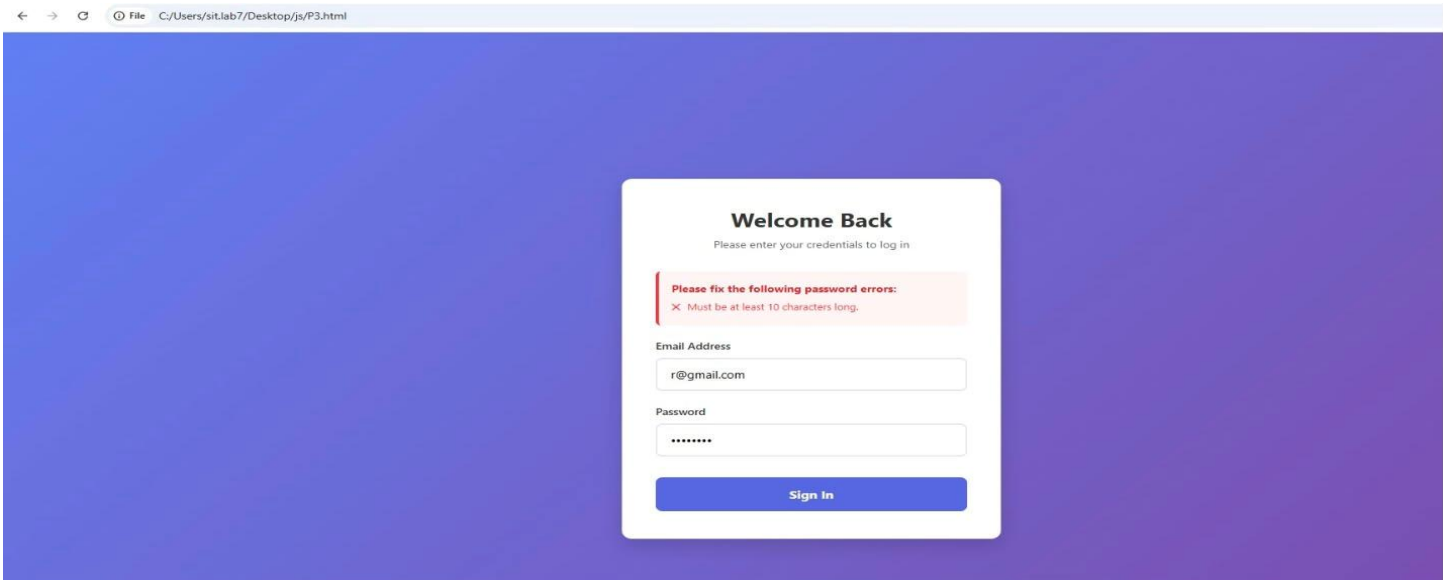
    var grade;
    var result;

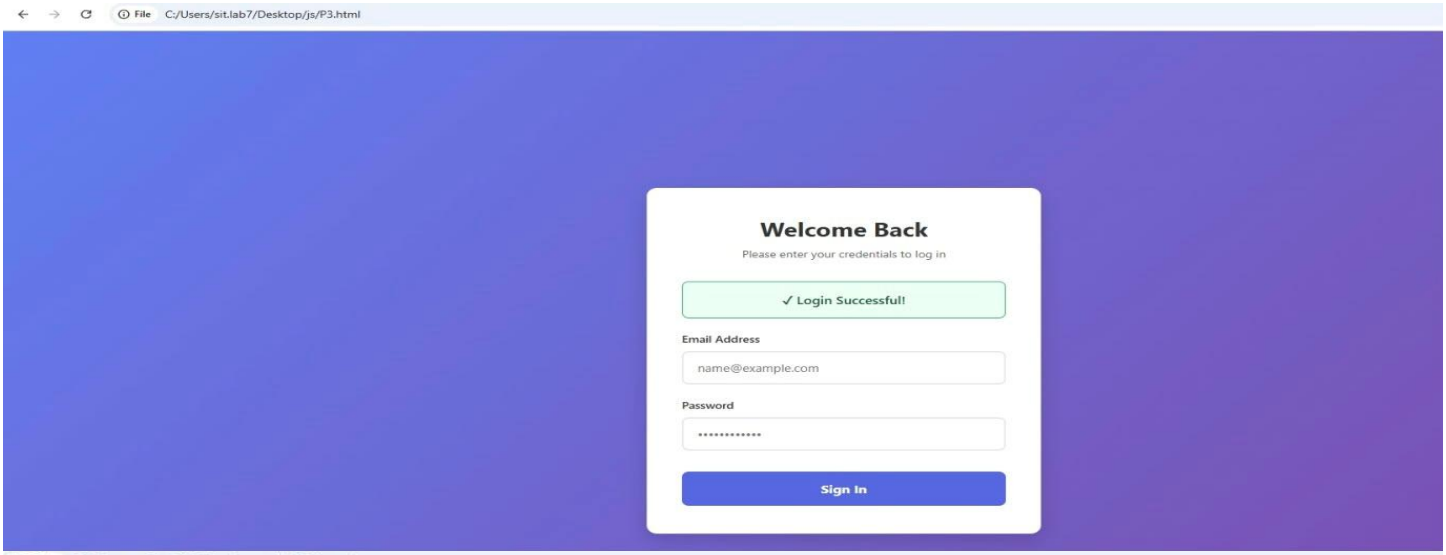
    if (marks >= 90) {
        grade = "A+";
        result = "Pass";
    }
    else if (marks >= 80) {
        grade  =  "A";
        result = "Pass";
    }
    else if (marks >= 70) {
        grade  =  "B";
        result = "Pass";
    }
    else if (marks >= 60) {
        grade  =  "C";
        result = "Pass";
    }
    else if (marks >= 50) {
        grade  =  "D";
        result = "Pass";
    }
    else if (marks >= 40) {
```

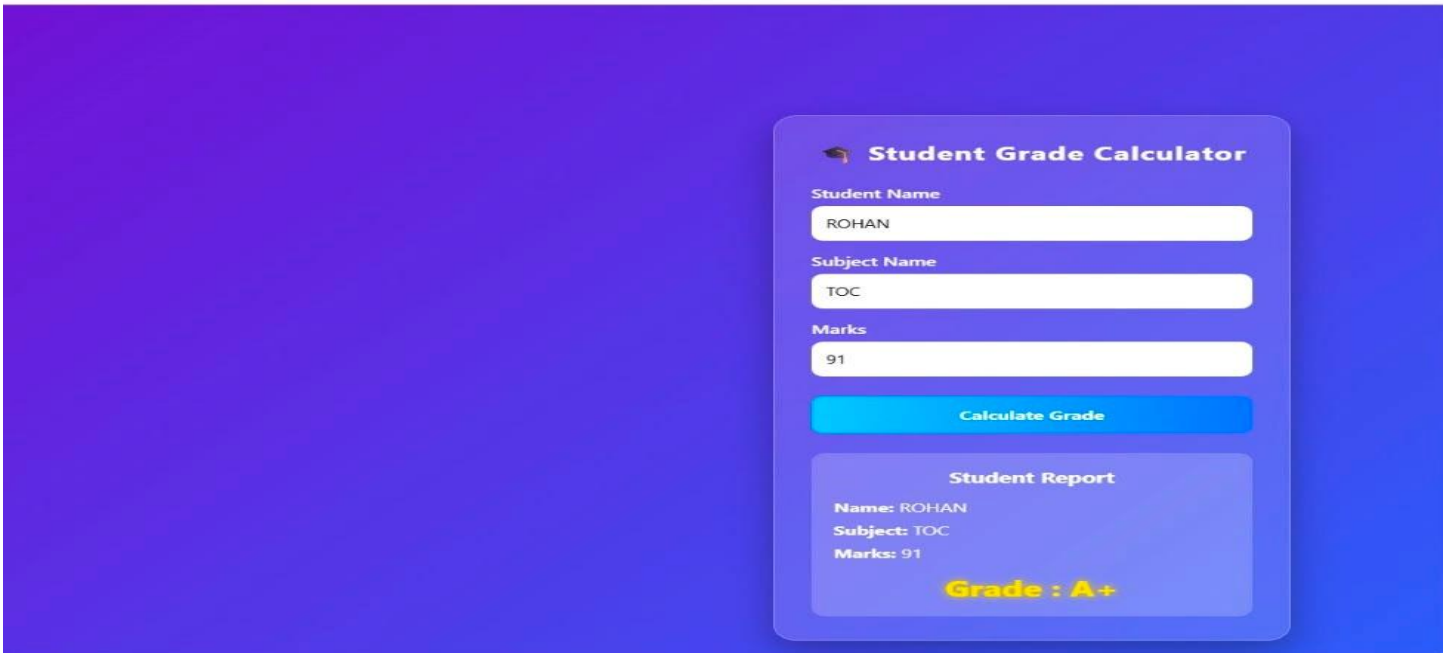
```
        grade = "E";
        result = "Pass";
    } else {
        grade = "F";
        result = "Fail";
    }

    var reportDiv = document.getElementById("report");
    if (reportDiv) {
        reportDiv.innerHTML = `
            <h2>Student Grade Report</h2>
            <hr>
            <div class="report-details">
                <p><b>Student Name:</b> <span>${name}</span></p>
                <p><b>Marks:</b> <span>${marks} / 100</span></p>
                <p><b>Grade:</b> <span class="badge grade-${grade.replace('+',
'plus')}">${grade}</span></p>
                <p><b>Result:</b> <span class="badge status-${result.toLowerCase()}">${result}</span></p>
            </div>
        `;
        reportDiv.classList.add("visible");
    } else {
        document.open();
        document.write(`
            <!DOCTYPE html>
            <html>
            <head>
                <title>Student Grade Report</title>
                <link rel="stylesheet" href="3.css">
            </head>
            <body class="report-view">
                <div class="container">
                    <div class="card result-card">
                        <h2>Student Grade Report</h2>
                        <hr>
                        <div class="report-details">
                            <p><b>Student Name:</b> <span>${name}</span></p>
                            <p><b>Marks:</b> <span>${marks} / 100</span></p>
                            <p><b>Grade:</b> <span class="badge grade-${grade.replace('+',
'plus')}">${grade}</span></p>
                            <p><b>Result:</b> <span class="badge status-
${result.toLowerCase()}">${result}</span></p>
                        </div>
                        <button class="btn btn-secondary" onclick="location.reload()">Back to
Form</button>
                    </div>
                </div>
            </body>
            </html>
        `);
        document.close();
    }
}
```

4. Output:


File C:/Users/sit.lab7/Desktop/js/P3.html
Welcome Back
Please enter your credentials to log in
Please fix the following password errors:
Must be at least 10 characters long.
Email Address
r@gmail.com
Password
Sign In


File C:/Users/sit.lab7/Desktop/js/P3.html
Welcome Back
Please enter your credentials to log in
✓ Login Successful!
Email Address
name@example.com
Password
Sign In


File C:/Users/sit.lab7/Desktop/js/p2.html
Student Grade Calculator
Student Name
ROHAN
Subject Name
TOC
Marks
91
Calculate Grade
Student Report
Name: ROHAN
Subject: TOC
Marks: 91
Grade : A+

5. Result/Conclusion:

Control structures (if-else if ladder) and form validation were successfully implemented to build a student grading system. The program validates that the name and marks fields are not empty and that marks lie between 0 and 100, then assigns a grade (A+ to F) and a Pass/Fail result based on the entered marks, displaying a formatted report on the page. This reinforced the use of conditional statements and input validation in JavaScript.